

Telco Customer Churn Prediction Using Machine Learning

1. Introduction

Customer churn prediction is an important problem in the telecommunications industry. Companies lose revenue when customers stop using their services. This project aims to analyze customer behavior and build a machine learning model capable of predicting whether a customer is likely to churn.

2. Dataset Collection

The dataset used in this project is the Telco Customer Churn dataset obtained from Kaggle/IBM.

The dataset contains customer demographic information, account details, subscribed services, and billing information. The target variable is "Churn", which indicates whether the customer left the company.

3. Data Cleaning & Preprocessing

Duplicate Removal

Duplicate rows were removed to ensure data quality and prevent biased learning.

Data Type Correction

The "TotalCharges" column contained inconsistent values and was converted into numeric format after removing invalid characters.

Missing Values Handling

Numerical missing values were filled using the median to reduce the impact of outliers, while categorical missing values were replaced with "Unknown".

Text Cleaning

Leading and trailing spaces were removed from text columns to improve consistency.

Pivot Tables

Pivot tables were used to analyze average monthly charges and customer distribution across churn categories and contract types.

4. Feature Extraction

Added Features

AvgMonthlySpend

Represents the customer's average spending behavior over subscription duration.

ServicesCount

Measures how many services the customer is subscribed to, which may reflect customer engagement.

TenureGroup

Converts tenure into behavioral categories for easier interpretation.

HighValue

Indicates whether the customer spends above the median monthly charges.

IsLongContract

Detects customers with long-term contracts who are expected to be more stable.

RiskScore

A custom risk indicator estimating churn likelihood based on contract type, monthly charges, and tenure.

5. Feature Selection

Correlation Analysis

Correlation analysis was used to measure the relationship between numerical features and churn.

Chi-Square Test

Chi-square testing was applied to categorical variables after encoding them numerically.

Random Forest Importance

Random Forest feature importance was used to identify the most influential predictors.

Comparison

Contract type, tenure, monthly charges, and risk-related features consistently appeared among the most important predictors across multiple selection methods.

6. Handling Data Imbalance

Problem Description

The dataset was imbalanced because churn customers represented a smaller percentage of the data.

Baseline Model

A baseline Random Forest model was trained before balancing to evaluate initial performance.

SMOTETomek

SMOTETomek was applied to combine oversampling and noise cleaning techniques.

Results

Model recall and F1-score improved significantly after balancing and threshold optimization.

7. Model Training & Optimization

Random Forest Model

Random Forest was selected because of its strong performance on structured tabular datasets and its ability to handle nonlinear relationships.

Hyperparameter Tuning

The following hyperparameters were tuned to improve model performance:

- `n_estimators`
- `max_depth`
- `min_samples_split`
- `min_samples_leaf`
- `class_weight`

Threshold Optimization

Different decision thresholds were tested to maximize churn detection while maintaining acceptable precision.

8. Results & Visualization

Churn Distribution

Most customers did not churn, indicating class imbalance.

Contract vs Churn

Month-to-month customers showed significantly higher churn rates.

Feature Importance

Tenure, contract type, and monthly charges were among the strongest predictors.

9. Conclusion

This project successfully applied data preprocessing, feature engineering, feature selection, imbalance handling, and machine learning techniques to predict customer churn. The final optimized model achieved improved recall and F1-score, making it more effective in identifying customers at risk of leaving.

10. Future Work

Future improvements may include testing deep learning models, adding customer interaction data, and deploying the model as a real-time prediction system.